

Vellacombe Discovery Ltd

Pyrrolopyridines: regioisomer separation and delivery

Route scout and delivery — final report, revision A

Programme: VD-96-072

Accession: VA19960654

Client: Elvarmere Biosciences AB

Date: 14 Feb 1997

Author: Dr R. Calderwood, Head of Medicinal Chemistry

1 Final material schedule

Elvarmere Biosciences AB received 24 compounds under VD-96-072. Analogue preparation began on 06 Dec 1996; the interim checkpoint was 09 Jan 1997, and chemistry completion was recorded on 13 Feb 1997. This final Route scout and delivery report is issued by Dr R. Calderwood on 14 Feb 1997.

Preparative separation secured the desired regioisomer and enabled the first comparative potency screen. This was the decisive route gate. The alternative regioisomer remained a route consequence, not an additional candidate to be included under the desired compound name. Hassan Darwish separated the question of making the core from that of recovering the assigned product; Ruth Calderwood retained the desired connectivity as the basis of the analogue set.

Table 1 brings the identifier, assigned structure, molecular weight, isolated yield, UV area purity and amount into the same entry. Only this schedule constitutes the reported delivery. Suggested structures elsewhere in the report are not additional isolated compounds.

Table 1. Compounds delivered; final isolated yield, HPLC purity in UV area percent at 254 nm and amount in mg.

Compound	SMILES	MW	Yield (%)	HPLC purity (UV area %, 254 nm)	Amount (mg)
EB05-08106	<chem>COc1ccc(-c2cnc3[nH]cc(C(=O)NCCN4CCN(CC4)c3c2)cc1</chem>	393.5	67	96.7	22
EB05-08110	<chem>CC(=O)Nc1ccc(-c2cnc3[nH]cc(C(=O)NCCN4CCC(O)CC4)c3c2)cc1</chem>	421.5	57	99.8	35
EB05-08128	<chem>CN1CCC(OCCNC(=O)c2c[nH]c3ncc(-c4ccc(C#N)cc4)cc23)CC1</chem>	403.5	31	97.2	6
EB05-08140	<chem>CN1CCN(CCNC(=O)c2c[nH]c3ncc(-c4cc(F)cc(F)c4)cc23)CC1</chem>	399.4	68	98.0	26
EB05-08152	<chem>O=C(NCCN1CC(C1)c1c[nH]c2ncc(-c3ccc(F)cc3)cc12</chem>	352.4	19	94.8	17
EB05-08158	<chem>CC(=O)Nc1ccc(-c2cnc3[nH]cc(C(=O)NCCN4CCN(CC4)c3c2)cc1</chem>	420.5	35	94.7	31
EB05-08172	<chem>COc1ccc(-c2cnc3[nH]cc(C(=O)NCCOC4CCN(CC4)c3c2)cc1</chem>	408.5	46	97.8	18
EB05-08178	<chem>NC1CCN(CCNC(=O)c2c[nH]c3ncc(-c4ccc(F)cc4)cc23)CC1</chem>	381.5	27	95.2	47
EB05-08182	<chem>CC(=O)Nc1ccc(-c2cnc3[nH]cc(C(=O)NCCN4CCOC(C4)c3c2)cc1</chem>	407.5	53	96.8	16
EB05-08188	<chem>Cc1ccc(-c2cnc3[nH]cc(C(=O)NCCNC4CCC(CC4)c3c2)nc1</chem>	377.5	44	98.1	31
EB05-08190	<chem>CN1CCC(OCCNC(=O)c2c[nH]c3ncc(-c4cc(F)cc(F)c4)cc23)CC1</chem>	414.5	56	96.3	28
EB05-08194	<chem>CN1CCN(CCNC(=O)c2c[nH]c3ncc(-c4ccc(F)cc4)cc23)CC1</chem>	381.5	28	95.6	42
EB05-08200	<chem>Cc1ccc(-c2cnc3[nH]cc(C(=O)NCCN4CCCC4)c3c2)nc1</chem>	349.4	33	95.7	5
EB05-08206	<chem>CCN1CCN(CCNC(=O)c2c[nH]c3ncc(-c4ccc(NC(C)=O)c</chem>	434.5	32	97.0	20

	<chem>c4)cc23)CC1</chem>				
EB05-08208	<chem>NC1CCN(CCNC(=O)c2c[nH]c3ncc(-c4cc(F)cc(F)c4)cc23)CC1</chem>	399.4	31	94.6	34
EB05-08218	<chem>CCN1CCN(CCN(C(=O)c2c[nH]c3ncc(-c4cncnc4)cc23)CC1</chem>	379.5	48	96.7	27
EB05-08230	<chem>Cc1ccc(-c2cnc3[nH]cc(C(=O)NCCN4CCOC4)c3c2)nc1</chem>	365.4	36	94.4	42
EB05-08236	<chem>CN1CCC(OCCN(C(=O)c2c[nH]c3ncc(-c4ccc(F)cc4)cc23)CC1</chem>	396.5	24	97.3	40
EB05-08248	<chem>N#Cc1ccc(-c2cnc3[nH]cc(C(=O)NCCNC4CCC4)c3c2)cc1</chem>	387.5	34	97.9	26
EB05-08250	<chem>CC(=O)Nc1ccc(-c2cnc3[nH]cc(C(=O)NCCN4CCCC4)c3c2)cc1</chem>	391.5	53	99.8	47
EB05-08256	<chem>Nc1ccc(-c2cnc3[nH]cc(C(=O)NCCN4CCCC4)c3c2)cc1</chem>	349.4	52	96.8	41
EB05-08278	<chem>O=C(NCCN1CC(C(O)CC1)c1c[nH]c2ncc(-c3ccc(F)cc3)cc12</chem>	382.4	28	97.4	42
EB05-08286	<chem>O=C(NCCN1CC(OCC1)c1c[nH]c2ncc(-c3ccc(F)cc3)cc12</chem>	368.4	4	97.4	18
EB05-08290	<chem>Nc1ccc(-c2cnc3[nH]cc(C(=O)NCCN4CCC(O)CC4)c3c2)cc1</chem>	379.5	50	94.3	55

2 Remit, chronology and responsibility

The remit was Analogue preparation for the EB05 set. The report covers route decisions affecting access to material and the identity basis needed for the client comparison. The programme turnaround was 69 days. That figure describes the programme interval, not a reaction time or assay incubation.

Dr R. Calderwood led the programme, Dr H. Darwish directed synthetic execution and Clara Fenwick led analytical release. The final gate closed on 13 Feb 1997; this report follows completion rather than reopening the route as a draft. Programme accession VA19960654 links the material schedule to the benchmark below.

Earlier intermediates remain useful experimental context but are not counted again as delivered compounds. A repeat request should name the released identifier and the chemistry question. The final schedule is the endpoint for reconciling the set; individual compound cards remain the source for recovering a preparation.

Table 2. Programme benchmark and administrative reconciliation.

Metric	Value
Project	VD-96-072
Client	Elvarmere Biosciences AB
Compounds tested	24
Series / chemotype	Elvarmere pyrrolopyridine series 05 (Pyrrolopyridine)
Assays	Custom synthesis and analogue preparation
Turnaround (calendar days)	69
Price	GBP 24,000

3 Chemistry hypothesis

The pyrrolopyridine programme required the desired ring connectivity to be carried into a comparative set without confusing it with the alternative regioisomer. The working hypothesis was that preparative separation would provide a more dependable route to the first comparative potency screen than continuing to pursue complete selectivity in formation of the core. A structure on a route sheet was not enough: the selected fraction had to become assigned, isolated material.

The programme was organised around material needed to answer the client question. Route success was judged at the isolated product, not at an attractive intermediate or plausible scheme. This distinction limits the final claims: chemistry establishes access and identity for the delivered set but supplies no biological ranking that has not been measured here. The proposed follow-up must retain the same distinction.

4 Route gate and sequence decision

Preparative separation secured the desired regioisomer and enabled the first comparative potency screen. This was the decisive route gate. The alternative regioisomer remained a route consequence, not an additional candidate to be included under the desired compound name. Hassan Darwish separated the question of making the core from that of recovering the assigned product; Ruth Calderwood retained the desired connectivity as the basis of the analogue set.

The interim discussion on 09 Jan 1997 was a route checkpoint, not the delivery endpoint. By 13 Feb 1997, the completed material schedule could be reconciled with the programme. A repeat chemist should preserve the reason for the route choice as well as the sequence. Repeating an operation without knowing which recovery or identity risk it controlled can recreate the original difficulty.

5 Analogue preparation

The assigned core was the basis for subsequent analogue preparation. The practical risk was carrying the alternative connectivity through a later transformation and mistaking a cleaner final chromatogram for proof of the original assignment. Each compound card must preserve the link to the separated precursor. A proposed substitution on the same drawing does not establish that the corresponding analogue has been made.

For repeat work, follow the compound card for the chosen identifier and reconcile the intended product with Table 1. A common scaffold is not a common experimental outcome. Substitution changes can affect work-up, purification and the evidence needed for assignment. No unprepared variant is claimed to have crossed the route gate merely because its drawing resembles a delivered compound.

6 Separation, work-up and recovery

Preparative fractions answer a recovery question; the regioisomer assignment answers an identity question. Mixed fractions cannot be counted as desired product merely because their nominal molecular mass agrees. Table 1 concerns the final reported compounds, not the crude core-forming mixture. Its isolated yields do not describe the regioselectivity of that reaction, and its amounts do not include a proposed recovery from overlapping fractions.

Isolated yield is a route measure and delivered amount is a material measure. Both matter because the comparison requires the actual compound in hand. Neither column tells a chemist where loss occurred. Return to intermediate and fraction entries for that question. An estimate derived only from the final table would conceal the difference between reaction performance and recovery.

7 Identity and assignment

The distinction between regioisomers depends on connectivity. Molecular mass alone cannot resolve it, and a dominant UV peak cannot establish which regioisomer is present. Clara Fenwick handled analytical release with the assignment carried from the route scout into the compound card. The structure field is a compact description of that assigned product, not a replacement for the supporting chemistry entry.

The release entry must stay connected to the isolate. A structure redrawn for a client presentation should retain the chemistry identifier so the assignment basis can be recovered. This report does not reproduce raw spectra or supply invented peak assignments. It records the programme conclusion and explains where the underlying experimental history is retained.

8 Purity statement and analytical limits

The HPLC values in Table 1 are UV area percent at 254 nm. They describe the reported chromatographic profile under that detection basis. They are not weight percent, solvent content or a direct assay of active parent in a weighed portion. The amount field is expressed in mg and must remain separate from purity when planning the comparison.

Clara Fenwick led analytical release. Identity and chromatographic purity answer complementary questions: a dominant peak cannot independently establish the assigned connection, position or configuration. An assigned structure also does not remove the need to consider impurities. The schedule retains the available numerical values without adding a universal purity threshold or an unstated correction to the delivered amounts.

A client query concerning precipitation, unexpected peaks or identity should return against the delivery identifier. The compact table is a reconciliation aid, not a substitute for the analytical entry needed to investigate a particular sample.

9 Isolated yield and material accounting

Final yields and amounts remain separated by compound. No pooled recovery or campaign mean has been calculated. Such a figure could blur route differences and imply a common experimental basis that the delivery table does not establish. The practical question is whether identifier, structure and amount refer to the same released material.

Compound cards may contain intermediate masses, recovered fractions and material held for further work. Those entries do not automatically belong to the client delivery. Use Table 1 for the delivered set and the route scout for an individual recovery question. A request for more material must distinguish repeat product from a new proposed analogue.

The programme closed with 24 delivered compounds. This count is not enlarged by an intermediate, an alternative drawing or a proposed comparison. Detailed weighing and transfer history belong to the individual chemistry entries; the final schedule supplies report-level reconciliation.

10 Client hand-off and identifier continuity

The EB05 identifiers connect isolated chemistry to the client comparison. Use VD-96-072 and accession VA19960654 for clarification. Assigned structures help reconcile a vial label with its intended product, but the delivery identifier remains the primary link to prepared material.

If a client transfers material for biological preparation, retain the original identifier and distinguish that later preparation description from chemistry release. Replacing an identifier with a proposed structure name risks merging different samples under a general scaffold label. The risk matters particularly where apparently similar structures address different chemical questions.

Chemistry completion was recorded on 13 Feb 1997. This report supplies no unstated courier date, receipt signature or storage condition. Dr H. Darwish is the route contact; Clara Fenwick is the analytical reconciliation contact; Dr R. Calderwood retains responsibility for the overall chemistry conclusion.

11 Interpretation for the client comparison

Elvarmere Biosciences AB can begin its first comparative potency screen with an assigned pyrrolopyridine set. This report supplies no cell-cycle kinase potency measurements and does not select a biological lead. Results should be returned against the delivered identifier. A response associated only with a proposed drawing would lose the link to the isolated material and could conceal the very regiochemical distinction addressed by the route.

The useful follow-up is a comparison whose members can be traced to released material. Keep the intended chemical difference explicit. Do not combine a proposed product and an isolated compound within a result entry. A later biological trend may justify new chemistry, but cannot change retrospectively which material was delivered.

This is a chemistry close-out. It provides no invented response values, selectivity ratios or potency rankings. Programme turnaround is not an assay setting, and isolated yield is not an indicator of target activity.

12 Next analogue and carried route risk

The next proposed analogue retains the established pyrrolopyridine connectivity and makes a single peripheral substitution change. It is outside this delivery. Keep the separated core as the route reference rather than changing the ring substitution and terminal amide together. If the new substituent changes separation behaviour, reassess the fraction boundary before reusing the previous collection pattern. Present route success does not make future regioisomeric mixtures equivalent.

A repeat route scout should identify the operation controlling the original risk and the evidence that closed it. Do not omit that gate because the final compound looks familiar. Substituent changes and material history can affect recovery without changing the intended chemical question. Preserve the link between the proposed route and the actual isolated outcome.

The report does not qualify a working scale beyond the delivered chemistry or impose new process conditions. Recover charge quantities, timings and purification parameters from the relevant experimental entry. A change in the established sequence needs its own reason and compound-card history.

13 Follow-up decision and status

The next analogue described above is proposed chemistry. It has no additional delivery entry, assigned yield, purity, amount or biological result in this report. Its purpose is a defined extension of the successful route while retaining the chemical distinction that motivated the programme.

Dr R. Calderwood recommends that Elvarmere Biosciences AB reconcile the intended comparison with the released identifiers before expanding preparation. Once the chemical question is selected, the next route scout can be scoped around it. A broad set assembled before that choice may add material without clarifying interpretation.

The gate for a new analogue remains identifiable isolated material. A convincing scheme is not a completed delivery. This final report closes the present programme and proposes a reasoned next step without presenting it as work already performed.

Appendix A — Delivery-table conventions

Compound is the delivery identifier linking material to chemistry. SMILES is the compact assigned-structure field. MW is the reported molecular-weight field associated with it. These fields must not be used to infer an unstated counter-ion, solvate or assignment detail. The drawing summarises identity; the compound card preserves its basis.

Yield (%) is the reported isolated yield. HPLC purity is UV area percent at 254 nm. Amount (mg) is the reported isolated amount. Keep the bases separate. The table supplies no raw peak areas, moisture correction, uncertainty estimate or across-programme normalisation. It also does not substitute for a preparation-specific experimental entry.

To reproduce a delivered product, select the identifier, recover its compound card and reconcile the final isolate with its route history. To extend the chemistry, create a route proposal rather than repurposing an existing delivery entry for an unmade structure.

Appendix B — Experimental history and repeat instructions

EB05_ISOL97.xls, closed on 13 Feb 1997, is the supporting route scout. It contains regioisomer assignments, isolated yields and delivery-lot purity. Use it with the compound cards to recover an individual preparation. The report does not invent charge weights or fraction volumes. A repeat entry must connect the core-forming operation, the separated fraction and the later transformations to the delivered vial.

A repeat request should identify the compound, why additional material is needed and the route question to preserve. The chemist can then recover the operation sequence and compare the repeat with the established identity basis. A broad scaffold description is insufficient when the conclusion depends on separation, selective access or assignment.

The report remains the final account of delivered chemistry. It does not overwrite experimental entries or create replacement conditions. If subsequent preparation departs from the established route, retain that departure in its own compound card and distinguish the new material history from the present delivery.

14 Final conclusion and authorisation

Preparative separation secured the desired regioisomer and enabled the first comparative potency screen. This was the decisive route gate. The alternative regioisomer remained a route consequence, not an additional candidate to be included under the desired compound name. Hassan Darwish separated the question of making the core from that of recovering the assigned product; Ruth Calderwood retained the desired connectivity as the basis of the analogue set.

The final delivery comprises 24 isolated compounds for Elvarmere Biosciences AB. Amounts, isolated yields and UV area purity are recorded in Table 1. The route supports the reported client comparison; proposed follow-up structures remain outside the delivery. No biological ranking is assigned in this conclusion.

Issued on 14 Feb 1997, after completion on 13 Feb 1997. Dr R. Calderwood authorises the chemistry conclusion. Dr H. Darwish is identified for synthetic execution and Clara Fenwick for analytical release. Revision A is final and has no open comments or provisional replacements for the material schedule.

Name	Role	Signature	Date
Dr R. Calderwood	Head of Medicinal Chemistry — final report authorisation	<u>D Calderwood</u>	14 Feb 1997